Autonomous Business Impact Monitoring System

Mentor :Soudip Roy Chowdhury, Yadunath Gupta, Amit D
Intern: Ryu Kakuta

Project Overview

We developed an Autonomous Business Impact Monitoring System using an agentic architecture that autonomously scrapes, filters, and analyzes global information streams with advanced NLP techniques. The system quantifies potential business impacts through multi-stage reasoning, combining real-time data extraction, scenario generation, and impact scoring. Built on a Redis-based streaming backend for scalable, low-latency processing, the system delivers actionable insights through a user-friendly, near real-time dashboard.

Background

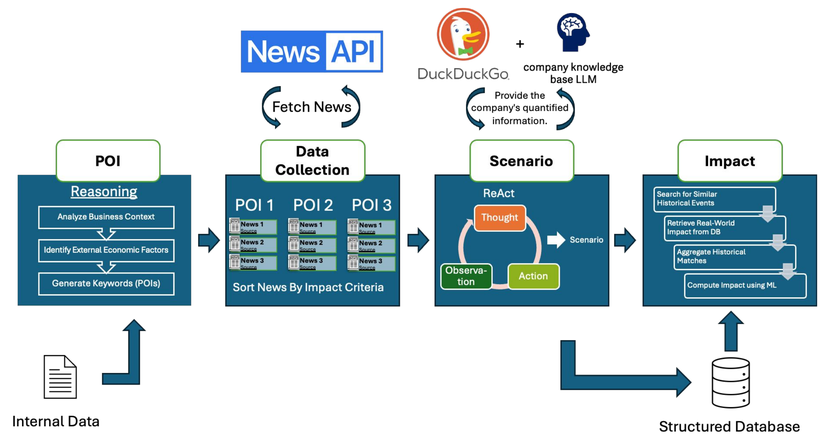
As supply chains span more regions, distant events, such as port delays or regulatory changes, can quickly disrupt local business operations. This research explores how automated systems can detect these risks early and provide timely, quantitative insights to support better decision-making.

Impact Monitoring Workflow

1. **POI Extraction Agent** – Uses LLMs and chain-of-thought prompting to extract domain-specific root-cause keywords from company data.
2. **Data Collection Agent** – Asynchronously queries NewsAPI.org to gather real, sourced articles matching each keyword.
3. **Scenario Generation Agent** – Uses the ReAct framework to draft plausible impact narratives, running ≥3 reasoning cycles per news item with LangChain agents, and DuckDuckGo search.
4. **Impact Quantification Agent** – Searches for historical events, retrieves real-world impact records from the database, aggregates historical matches, and applies forecasting models to compute projected impacts.

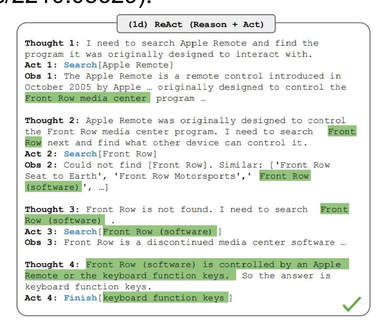
Orchestrated with StateGraph and Redis Streams, the system coordinates multi-agent workflows to deliver low-latency, real-time alerts and quantitative impact forecasts, helping researchers quickly translate global disruptions into actionable insights on supply chains, sales, and operations.

System Architecture



ReAct (Reasoning + Acting) Prompting

ReAct is a prompting technique that enables language models to reason step-by-step while interacting with external tools to solve complex problems (https://arxiv.org/abs/2210.03629).



(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

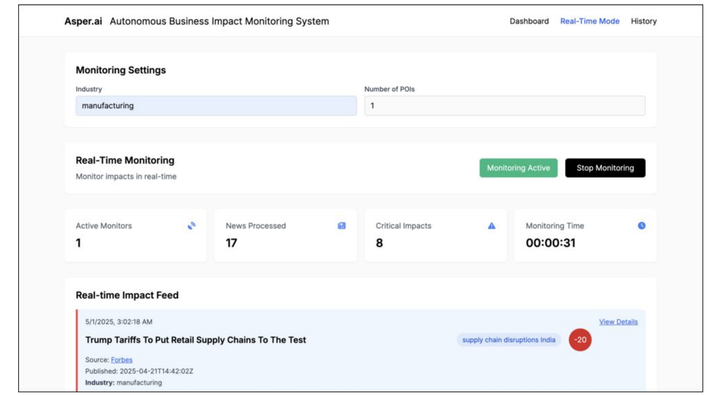
Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports','Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]

In our system, ReAct allows the LLM to generate realistic business impact scenarios by combining external news searches (via DuckDuckGo) with internal company-specific knowledge (through a company knowledge base LLM). This approach enables both contextual understanding and quantitative assessment of how real-world events may affect business operations.

User Interface



Future Implementation

- **Enhanced Search & Retrieval**: Use Elasticsearch and vector databases (e.g., FAISS, Pinecone) for faster, semantically accurate querying of past incidents.
- **Smarter Scenario Modeling**: Separate constant and variable components in scenarios to compute impacts more precisely.
- **Adaptive Historical Analysis**: Dynamically adjust time windows and apply semantic clustering with embedding models to improve relevance and reduce redundancy.

tech stack

LangGraph, mongoDB, Flask, tmux, CELERY, redis

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